6.5 Properties of Logarithms

Name _____ Period ____ Date _____

Algebra II 6.5 · Student sheet

I can…

- use the properties of logarithms to evaluate logarithms.
- expand and condense logarithmic expressions.
- use the change-of-base formula.

Part 1 · Calculator test

Use a calculator. Round to three decimal places. Are the two columns equal?

Expression	Value	Compare with	Value	Equal?
log 2 + log 3		log 6		
log 12 − log 4		log 3		
3 log 2		log 8		
log 5 + log 5		log 10		
log 10 − log 2		log 8		

Rules: log m + log n = _____ log m − log n = _____ n log m = _____

Part 2 · Notes: fill in the blanks

Product Property: $\log_b mn$ = _____

Quotient Property: $\log_b (m/n)$ = _____

Power Property: $\log_b m^n$ = _____

Change of base: $\log_c a$ = _____ = _____

Part 3 · Examples (work them with the class)

Example 1

Use $\log_2 3 \approx 1.585$ and $\log_2 7 \approx 2.807$ to evaluate each logarithm.

a. $\log_2 21$ b. $\log_2 \left(\frac{7}{3}\right)$ c. $\log_2 49$

Example 2

Expand $\ln \left(\frac{5x^7}{y}\right)$.

Example 3

Expand $\log_3 (x^2\sqrt{y})$.

Example 4

Condense $\log 9 + 3 \log 2 - \log 3$.

Example 5

Evaluate $\log_3 8$ using common logarithms and natural logarithms.

Example 6

The loudness of a sound is $L(I) = 10 \log (\frac{I}{I_0})$, where I is its intensity. How much louder (in decibels) is a sound when its intensity is doubled?

Part 4 · Partner practice

Partner 1

Expand $\log_6 (3x^4)$.

Partner 2

Condense $2 \ln x + \ln 5 - \ln y$.